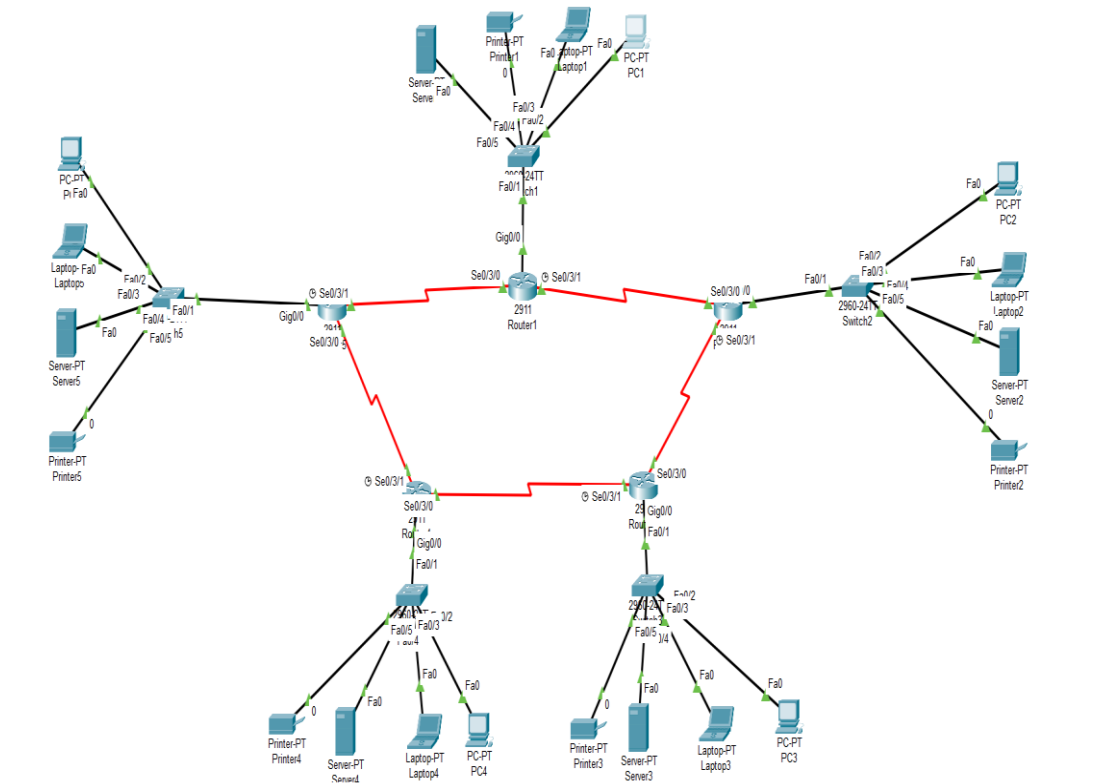


CCNA Capstone Project: Global Logistics Network

1-Network Topology

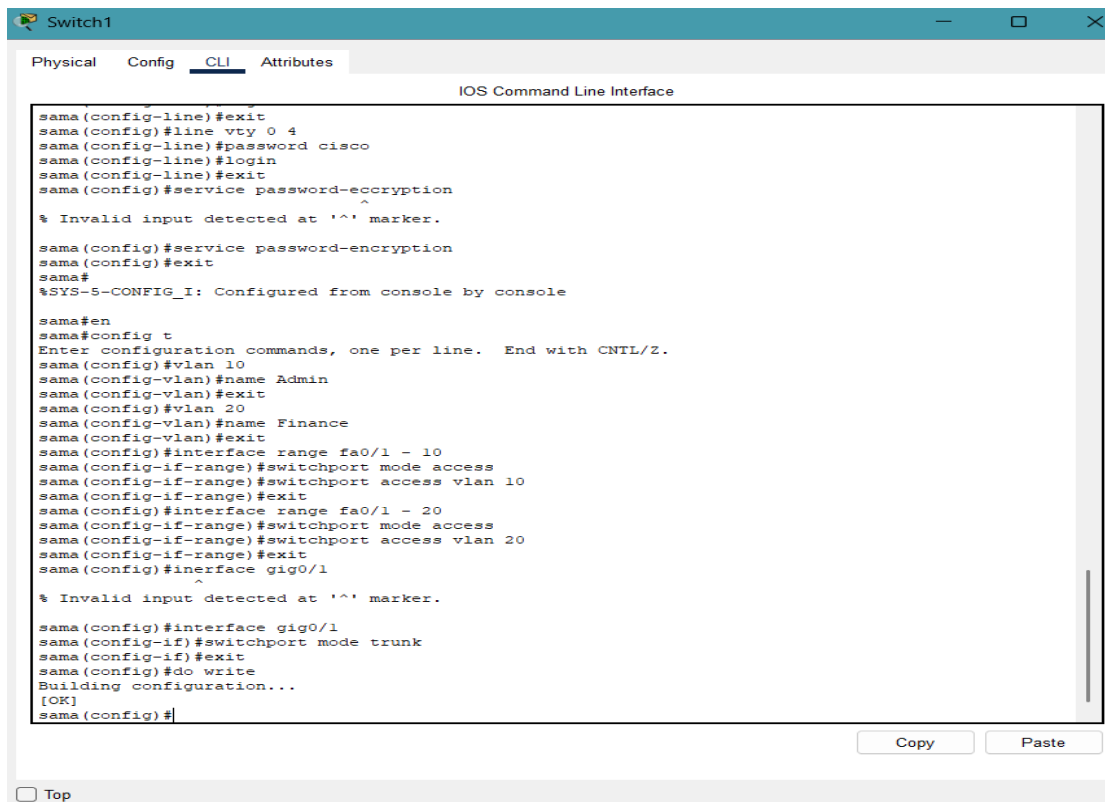
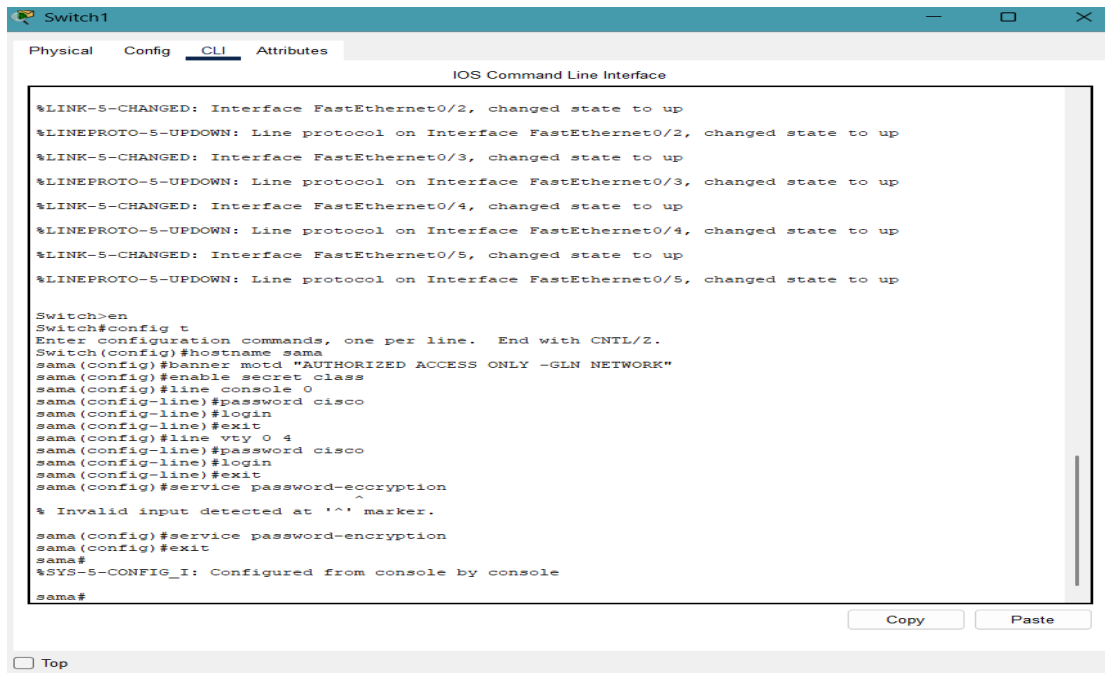
This topology represents a redundant ring network connecting 5 Branches using EIGRP as the dynamic routing protocol.



2-Switch configurations:

In this section, I performed the initial device hardening. This includes setting a hostname, securing the console and remote access with passwords, and enabling global password encryption to protect sensitive data.

I implemented VLAN (Virtual Local Area Network) technology to logically segment the network into different departments. This improves security and reduces broadcast traffic, ensuring that each department stays within its own broadcast domain.



Switch2

Physical Config CLI Attributes

IOS Command Line Interface

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname sama
sama(config)#banner motd "AUTHORIZED ACCESS ONLY -GLN NETWORK"
sama(config)#enable secret class
sama(config)#line console 0
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#line vty 0 4
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#service password-encryption
sama(config)#exit
sama#
%SYS-5-CONFIG_I: Configured from console by console
sama#

☐ Top

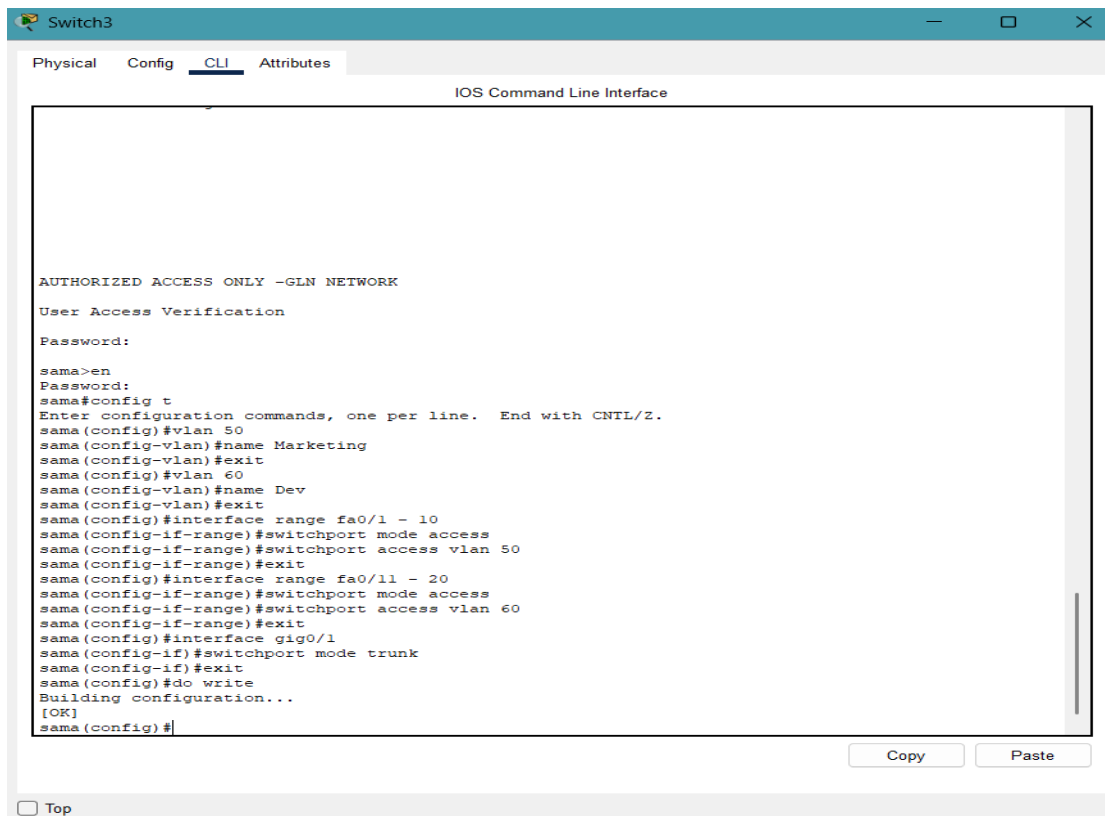
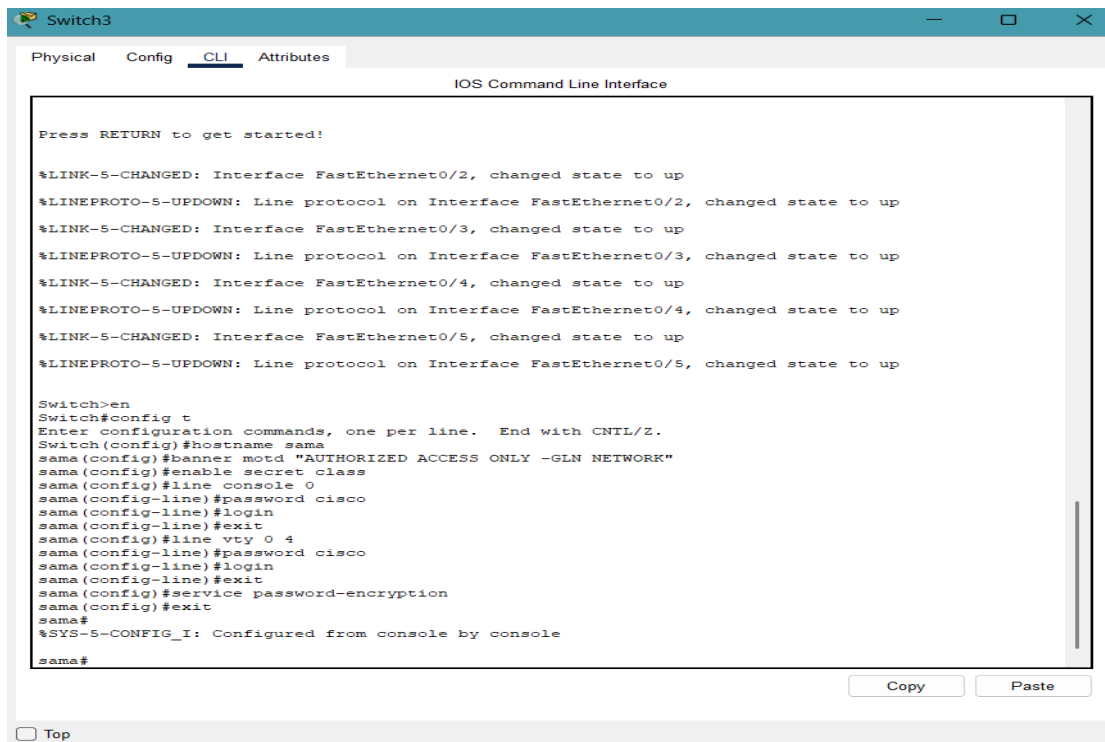
Switch2

Physical Config CLI Attributes

IOS Command Line Interface

Switch(config)#hostname sama
sama(config)#banner motd "AUTHORIZED ACCESS ONLY -GLN NETWORK"
sama(config)#enable secret class
sama(config)#line console 0
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#line vty 0 4
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#service password-encryption
sama(config)#exit
sama#
%SYS-5-CONFIG_I: Configured from console by console
sama#en
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#vlan 30
sama(config-vlan)#name Sales
sama(config-vlan)#exit
sama(config)#vlan 40
sama(config-vlan)#name Support
sama(config-vlan)#exit
sama(config)#interface range fa0/1 - 10
sama(config-if-range)#switchport mode access
sama(config-if-range)#switchport access vlan 30
sama(config-if-range)#exit
sama(config)#interface range fa0/1 - 20
sama(config-if-range)#switchport mode access
sama(config-if-range)#switchport access vlan 40
sama(config-if-range)#exit
sama(config)#interface gig0/1
sama(config-if)#switchport trunk
% Incomplete command.
sama(config-if)#switchport mode trunk
sama(config-if)#exit
sama(config)#do write
Building configuration...
[OK]
sama(config)#

☐ Top



Switch4

Physical Config CLI Attributes

IOS Command Line Interface

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname sama
sama(config)#banner motd "AUTHORIZED ACCESS ONLY --GLN NETWORK"
sama(config)#line console 0
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#line vty 0 4
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#service password-encryption
sama(config)#exit
sama#
%SYS-5-CONFIG_I: Configured from console by console
sama#

Copy Paste

☐ Top

Switch4

Physical Config CLI Attributes

IOS Command Line Interface

Press RETURN to get started.

AUTHORIZED ACCESS ONLY --GLN NETWORK

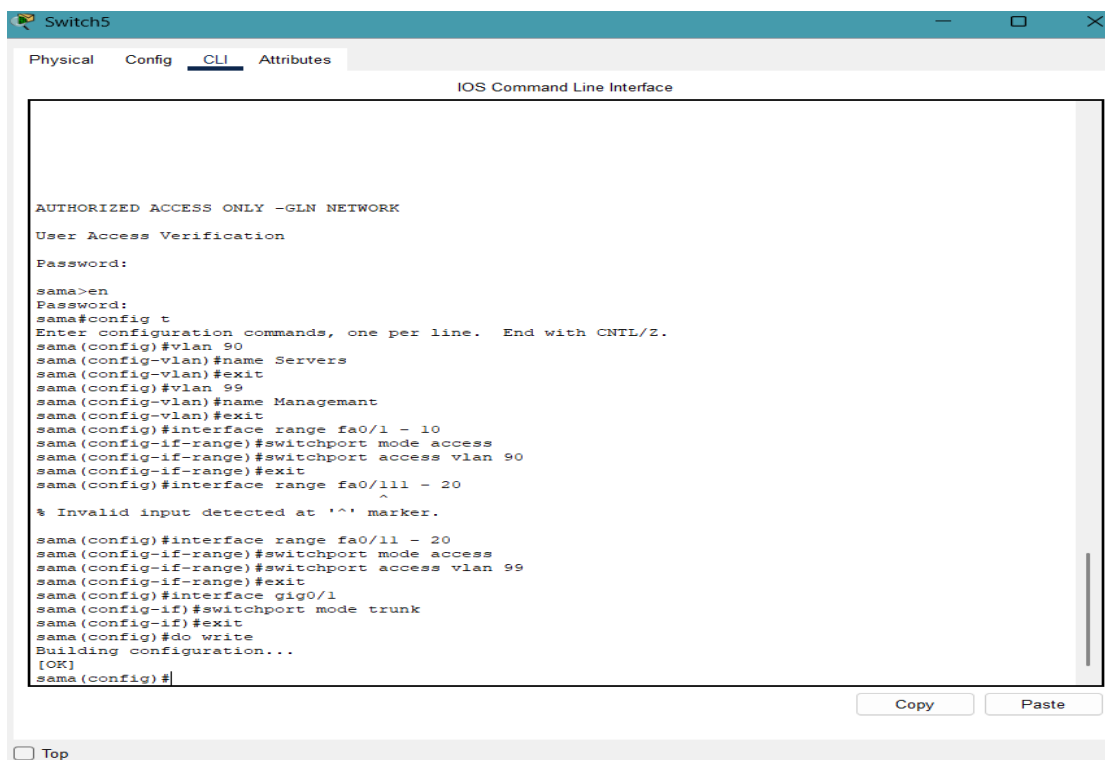
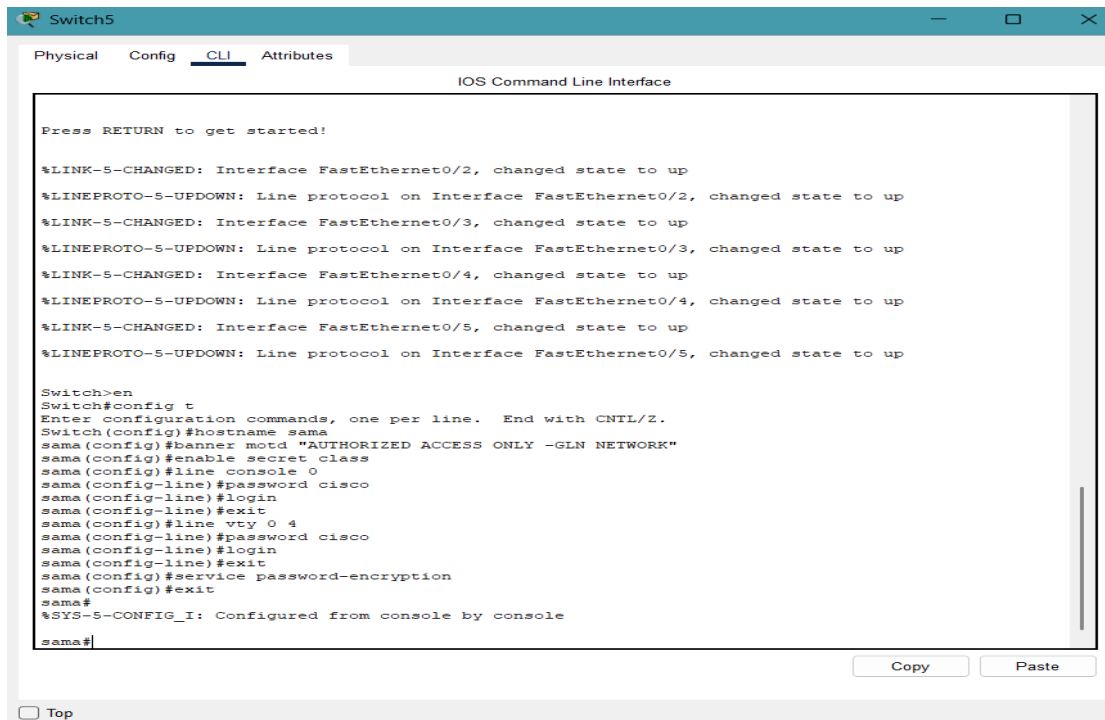
User Access Verification

Password:

sama>en
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#vlan 70
sama(config-vlan)#name Inventory
sama(config-vlan)#exit
sama(config)#vlan 80
sama(config-vlan)#name Shipping
sama(config-vlan)#exit
sama(config)#interface range fa0/1 - 10
sama(config-if-range)#switchport mode access
sama(config-if-range)#switchport access vlan 70
sama(config-if-range)#exit
sama(config)#interface range fa0/11 - 20
sama(config-if-range)#switchport mode access
sama(config-if-range)#switchport access vlan 80
sama(config-if-range)#exit
sama(config)#interface gig0/1
sama(config-if)#switchport mode trunk
sama(config-if)#exit
sama(config)#do write
Building configuration...
[OK]
sama(config)#

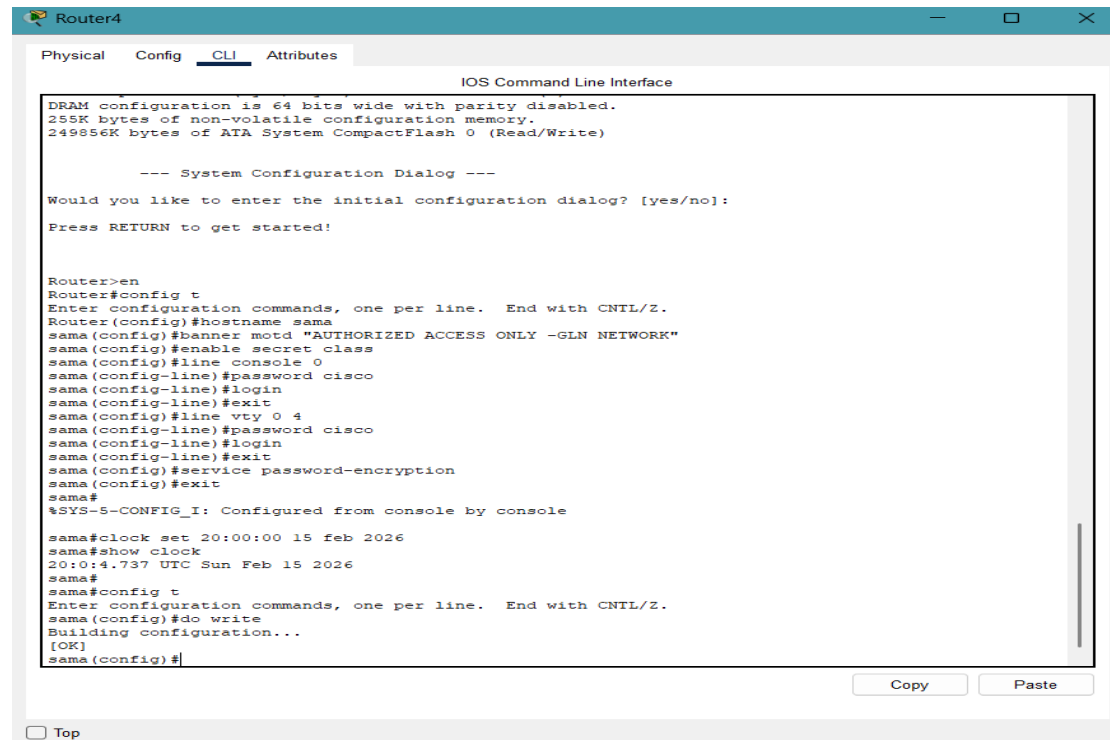
Copy Paste

☐ Top



3-Router Configurations:

In this section, I performed the initial device hardening. This includes setting a hostname, securing the console and remote access with passwords, and enabling global password encryption to protect sensitive data.



The screenshot shows the CLI of Router4. It starts with a system configuration dialog asking if the user wants to enter the initial configuration. The user responds 'no'. Then, the user enters 'en' to enter configuration mode. The user then enters 'config t' to enter global configuration mode. The user then enters 'hostname sama' to set the hostname. The user then enters 'banner motd "AUTHORIZED ACCESS ONLY -GLN NETWORK"' to set the MOTD banner. The user then enters 'enable secret class' to set the enable password. The user then enters 'line console 0' to enter console line configuration. The user then enters 'password cisco' to set the console password. The user then enters 'login' to enable console login. The user then enters 'exit' to exit console line configuration. The user then enters 'line vty 0 4' to enter vty line configuration. The user then enters 'password cisco' to set the vty password. The user then enters 'login' to enable vty login. The user then enters 'exit' to exit vty line configuration. The user then enters 'service password-encryption' to enable global password encryption. The user then enters 'exit' to exit global configuration mode. The user then enters 'show' to display the current configuration. The output shows the hostname 'sama', the MOTD banner, the enable password 'class', and the vty password 'cisco'. The user then enters 'copy' to save the configuration to the startup configuration. The output shows the configuration being saved to the startup configuration.

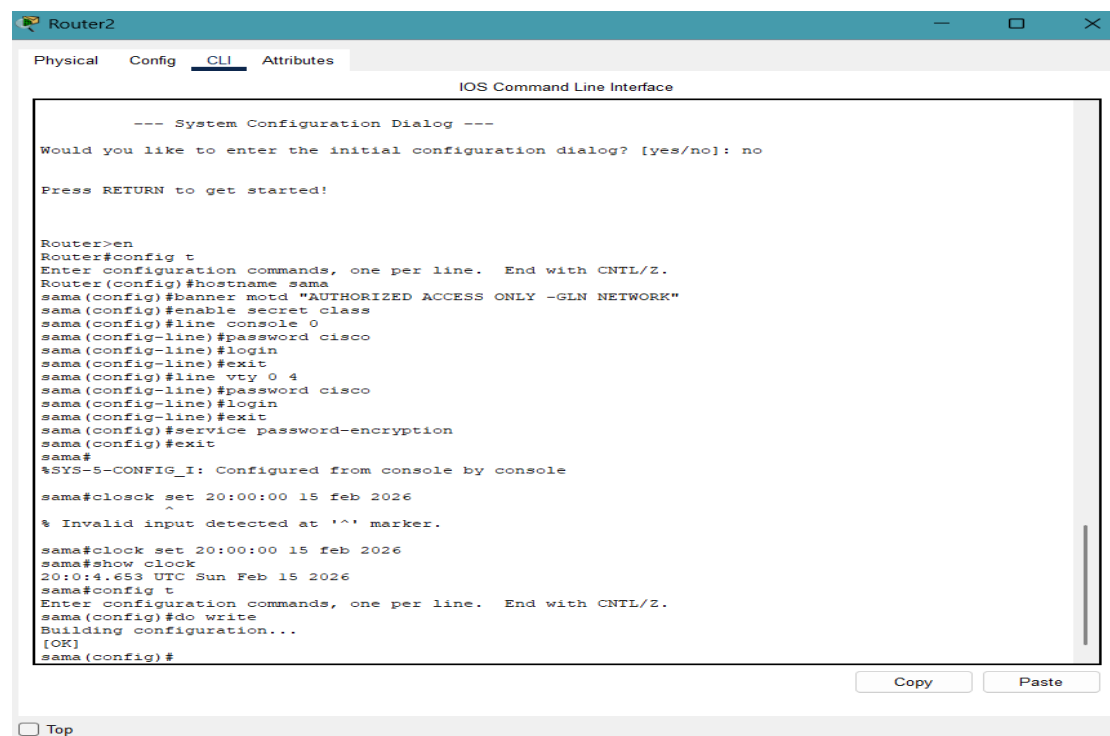
```
Router4
Physical Config CLI Attributes
IOS Command Line Interface

DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]:
Press RETURN to get started!

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname sama
sama(config)#banner motd "AUTHORIZED ACCESS ONLY -GLN NETWORK"
sama(config)#enable secret class
sama(config)#line console 0
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#line vty 0 4
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#service password-encryption
sama(config)#exit
sama#
%SYS-5-CONFIG_I: Configured from console by console

sama#clock set 20:00:00 15 feb 2026
sama#show clock
20:00:4.737 UTC Sun Feb 15 2026
sama#
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#do write
Building configuration...
[OK]
sama(config)#
```



The screenshot shows the CLI of Router2. It starts with a system configuration dialog asking if the user wants to enter the initial configuration. The user responds 'no'. Then, the user enters 'en' to enter configuration mode. The user then enters 'config t' to enter global configuration mode. The user then enters 'hostname sama' to set the hostname. The user then enters 'banner motd "AUTHORIZED ACCESS ONLY -GLN NETWORK"' to set the MOTD banner. The user then enters 'enable secret class' to set the enable password. The user then enters 'line console 0' to enter console line configuration. The user then enters 'password cisco' to set the console password. The user then enters 'login' to enable console login. The user then enters 'exit' to exit console line configuration. The user then enters 'line vty 0 4' to enter vty line configuration. The user then enters 'password cisco' to set the vty password. The user then enters 'login' to enable vty login. The user then enters 'exit' to exit vty line configuration. The user then enters 'service password-encryption' to enable global password encryption. The user then enters 'exit' to exit global configuration mode. The user then enters 'show' to display the current configuration. The output shows the hostname 'sama', the MOTD banner, the enable password 'class', and the vty password 'cisco'. The user then enters 'copy' to save the configuration to the startup configuration. The output shows the configuration being saved to the startup configuration.

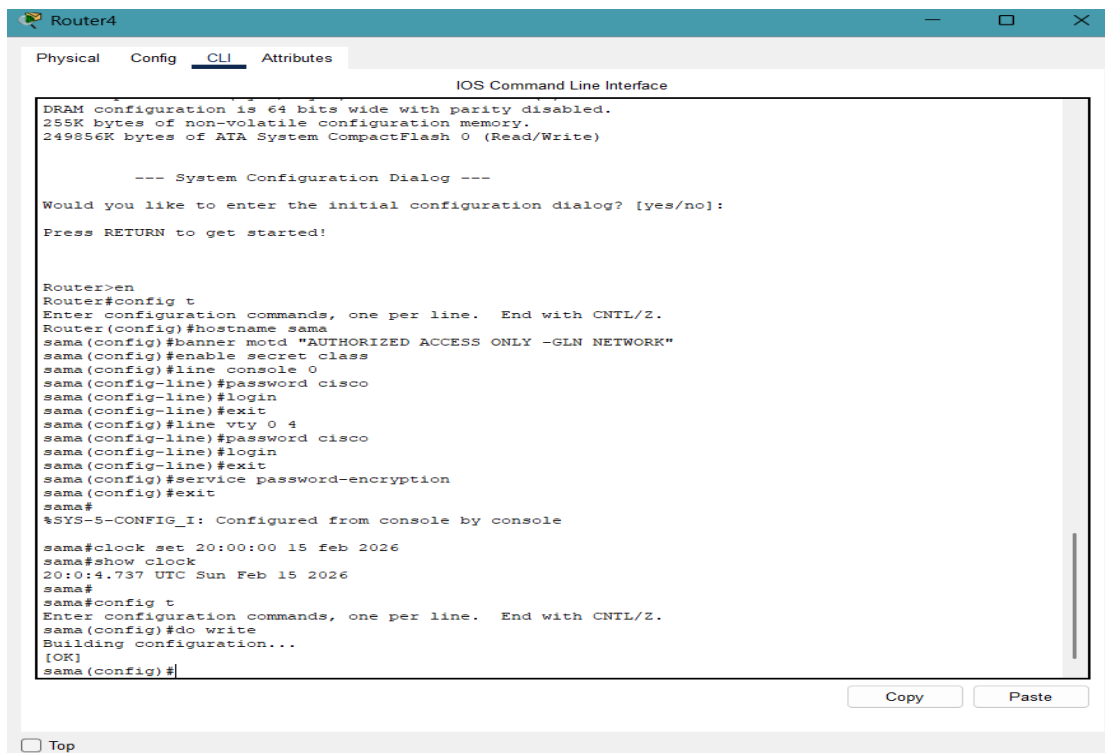
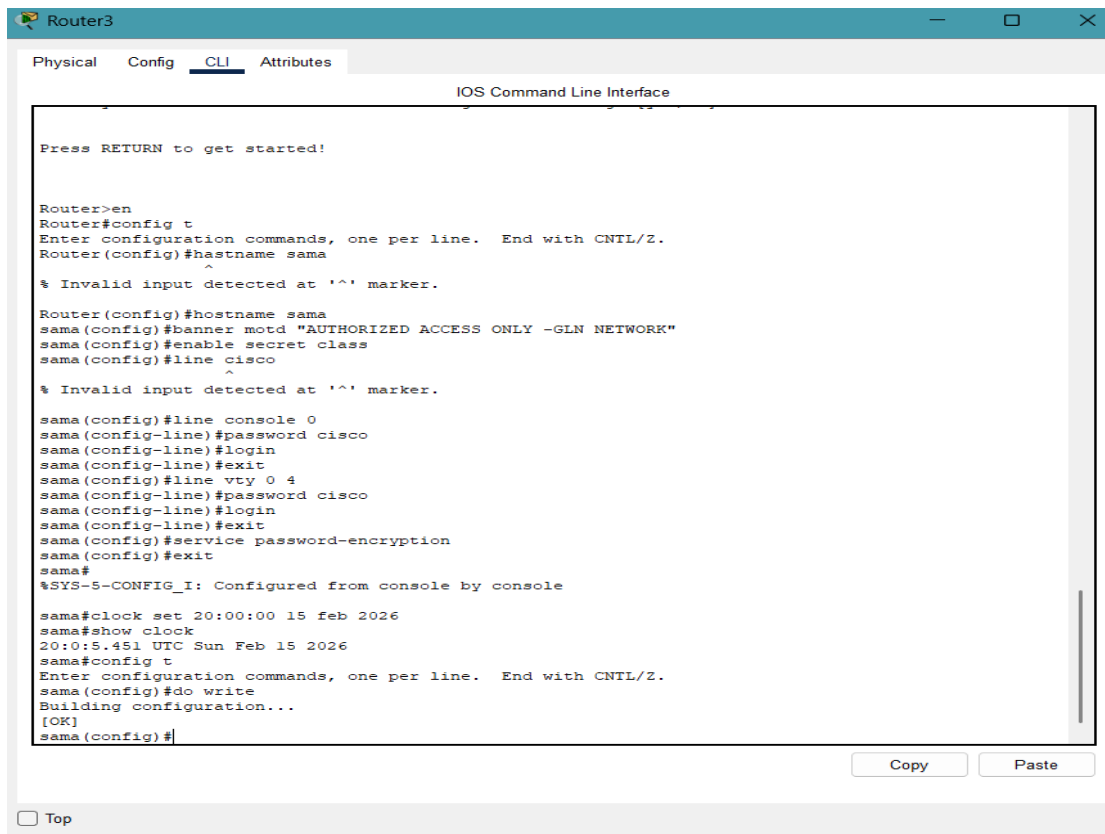
```
Router2
Physical Config CLI Attributes
IOS Command Line Interface

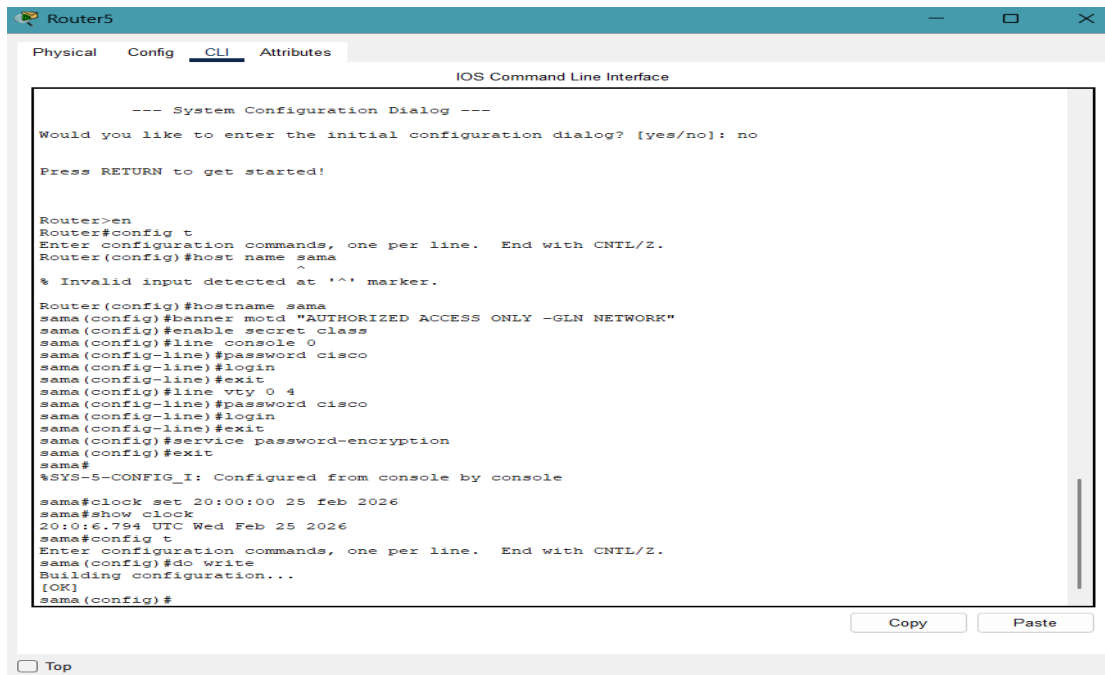
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no
Press RETURN to get started!

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname sama
sama(config)#banner motd "AUTHORIZED ACCESS ONLY -GLN NETWORK"
sama(config)#enable secret class
sama(config)#line console 0
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#line vty 0 4
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#service password-encryption
sama(config)#exit
sama#
%SYS-5-CONFIG_I: Configured from console by console

sama#clock set 20:00:00 15 feb 2026
% Invalid input detected at '^' marker.

sama#clock set 20:00:00 15 feb 2026
sama#show clock
20:00:4.653 UTC Sun Feb 15 2026
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#do write
Building configuration...
[OK]
sama(config)#
```





The screenshot shows the CLI window for Router5. The window has tabs for Physical, Config, CLI (selected), and Attributes. The title bar says "Router5". The main area is titled "IOS Command Line Interface". It displays the "System Configuration Dialog" where the user has chosen "no" to enter the initial configuration. The user then enters "en" to enter configuration mode. The configuration commands entered are: "config t", "hostname sama", "banner motd 'AUTHORIZED ACCESS ONLY -GLN NETWORK'", "enable secret class", "line console 0", "password cisco", "login", "exit", "line vty 0 4", "password cisco", "login", "exit", "service password-encryption", "exit", "do write", and "clock set 20:00:00 25 feb 2026". The window also shows the current time and date, and a "Top" button at the bottom left.

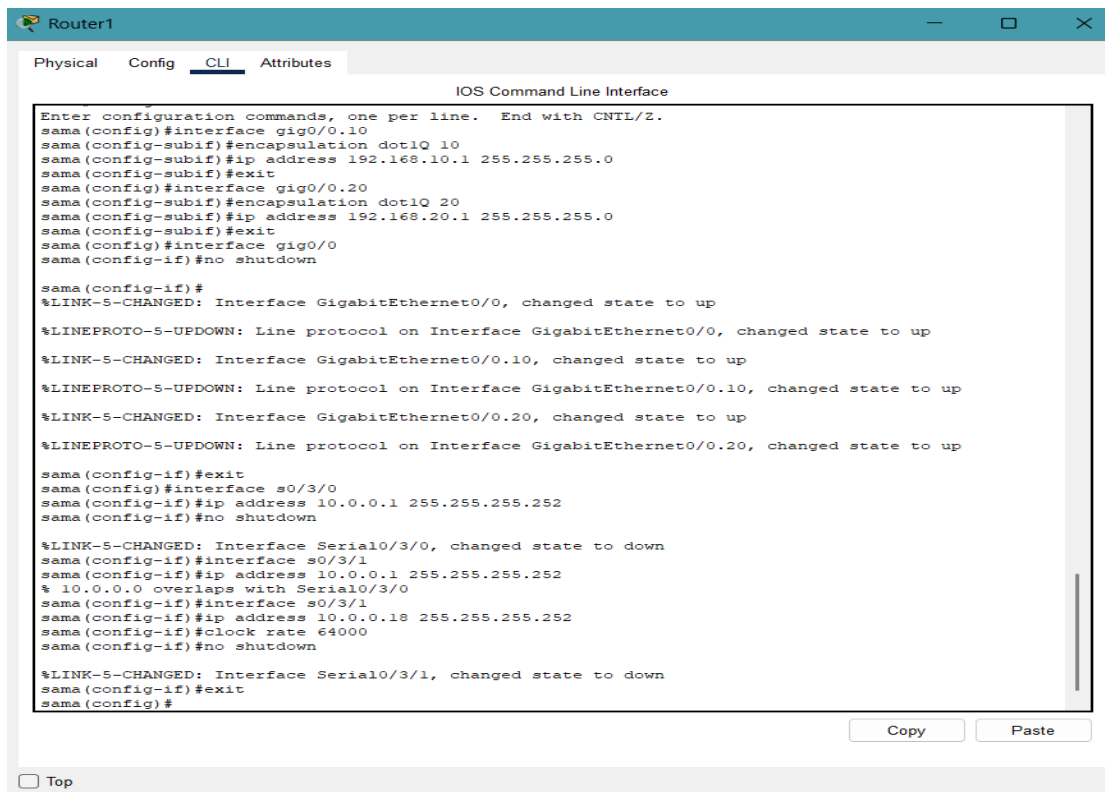
```
Router5
Physical Config CLI Attributes
IOS Command Line Interface

--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no
Press RETURN to get started!

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname sama
Router(config)#banner motd "AUTHORIZED ACCESS ONLY -GLN NETWORK"
% Invalid input detected at '^' marker.
Router(config)#hostname sama
sama(config)#enable secret class
sama(config)#line console 0
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#line vty 0 4
sama(config-line)#password cisco
sama(config-line)#login
sama(config-line)#exit
sama(config)#service password-encryption
sama(config)#exit
sama#
%SYS-5-CONFIG_I: Configured from console by console

sama#clock set 20:00:00 25 feb 2026
sama#show clock
20:00:06.794 UTC Wed Feb 25 2026
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#do write
Building configuration...
[OK]
sama(config)#
```

After creating the VLANs, I assigned specific ports to their respective VLANs in Access mode. Additionally, I configured the Gigabit ports as Trunk links to allow multiple VLANs to communicate across the entire network infrastructure.



The screenshot shows the CLI window for Router1. The window has tabs for Physical, Config, CLI (selected), and Attributes. The title bar says "Router1". The main area is titled "IOS Command Line Interface". It displays the configuration commands entered: "interface gig0/0.10", "encapsulation dot1Q 10", "ip address 192.168.10.1 255.255.255.0", "exit", "interface gig0/0.20", "encapsulation dot1Q 20", "ip address 192.168.20.1 255.255.255.0", "exit", "interface gig0/0", "no shutdown", "exit", "interface s0/3/0", "ip address 10.0.0.1 255.255.255.252", "no shutdown", "exit", "interface s0/3/1", "ip address 10.0.0.18 255.255.255.252", "clock rate 64000", "no shutdown", "exit", and "exit". The window also shows the current time and date, and a "Top" button at the bottom left.

```
Router1
Physical Config CLI Attributes
IOS Command Line Interface

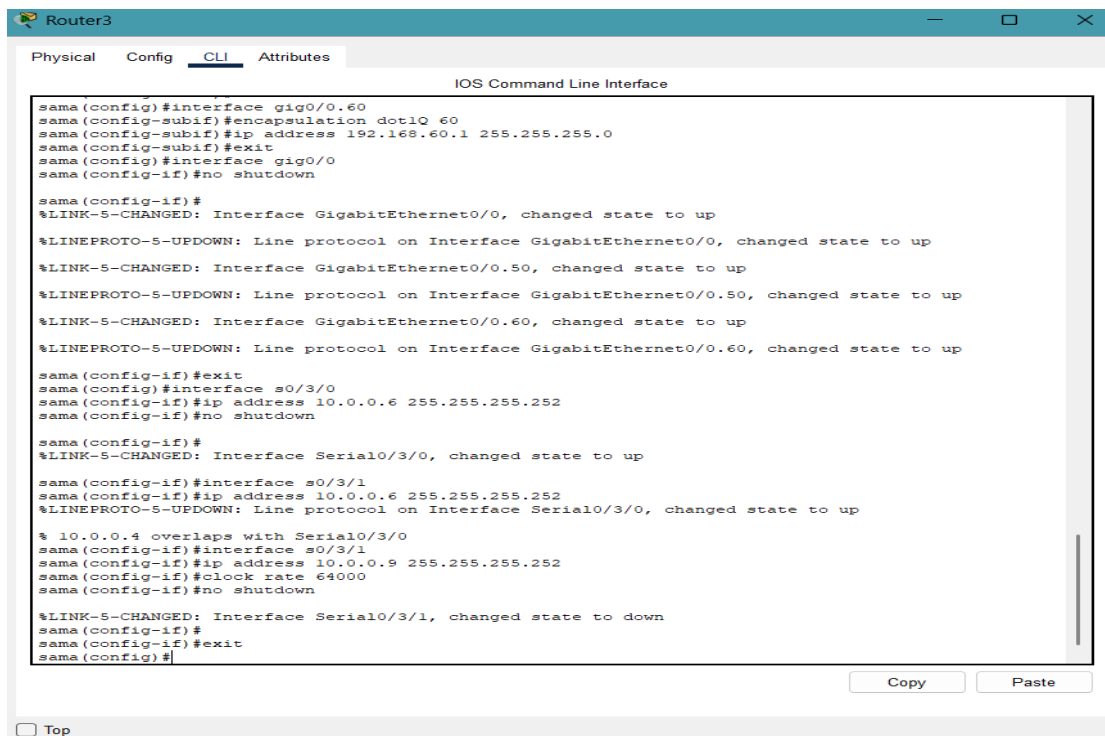
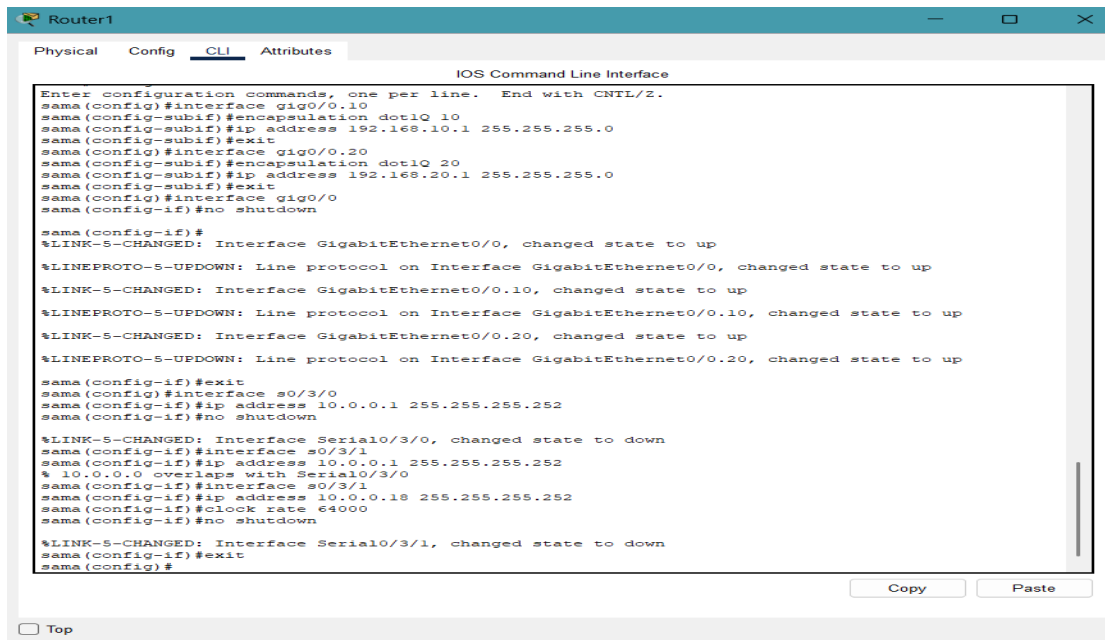
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#interface gig0/0.10
sama(config-subif)#encapsulation dot1Q 10
sama(config-subif)#ip address 192.168.10.1 255.255.255.0
sama(config-subif)#exit
sama(config)#interface gig0/0.20
sama(config-subif)#encapsulation dot1Q 20
sama(config-subif)#ip address 192.168.20.1 255.255.255.0
sama(config-subif)#exit
sama(config)#interface gig0/0
sama(config-if)#no shutdown

sama(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

sama(config-if)#exit
sama(config)#interface s0/3/0
sama(config-if)#ip address 10.0.0.1 255.255.255.252
sama(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
sama(config-if)#interface s0/3/1
sama(config-if)#ip address 10.0.0.18 255.255.255.252
% 10.0.0.0 overlaps with Serial0/3/0
sama(config-if)#clock rate 64000
sama(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
sama(config-if)#exit
sama(config)#
```



Router4

Physical Config **CLI** Attributes

IOS Command Line Interface

```
sama(config)#interface gig0/0.80
sama(config-subif)#encapsulation dot1Q 80
sama(config-subif)#ip address 192.168.80.1 255.255.255.0
sama(config-subif)#exit
sama(config)#interface gig0/0
sama(config-if)#no shutdown

sama(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0.70, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.70, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/0.80, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.80, changed state to up

sama(config-if)#exit
sama(config)#interface 0/3/0
      ^
% Invalid input detected at '^' marker.

sama(config)#interface s0/3/0
sama(config-if)#ip address 10.0.0.10 255.255.255.252
sama(config-if)#no shutdown

sama(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

sama(config-if)#interface s0/3/1
sama(config-if)#interface s0/3/1
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, chan
sama(config-if)#ip address 10.0.0.13 255.255.255.252
sama(config-if)#clock rate 64000
sama(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
sama(config-if)#exit
sama(config)#
```

☐ Top

Router5

Physical Config **CLI** Attributes

IOS Command Line Interface

```
sama(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0.90, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.90, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/0.99, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.99, changed state to up

sama(config-if)#exit
sama(config)#interface s0/3/0
      ^
% Invalid input detected at '^' marker.

sama(config)#interface s0/3/0
sama(config-if)#ip address 10.0.0.14 255.255.255.252
sama(config-if)#no shutdown

sama(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

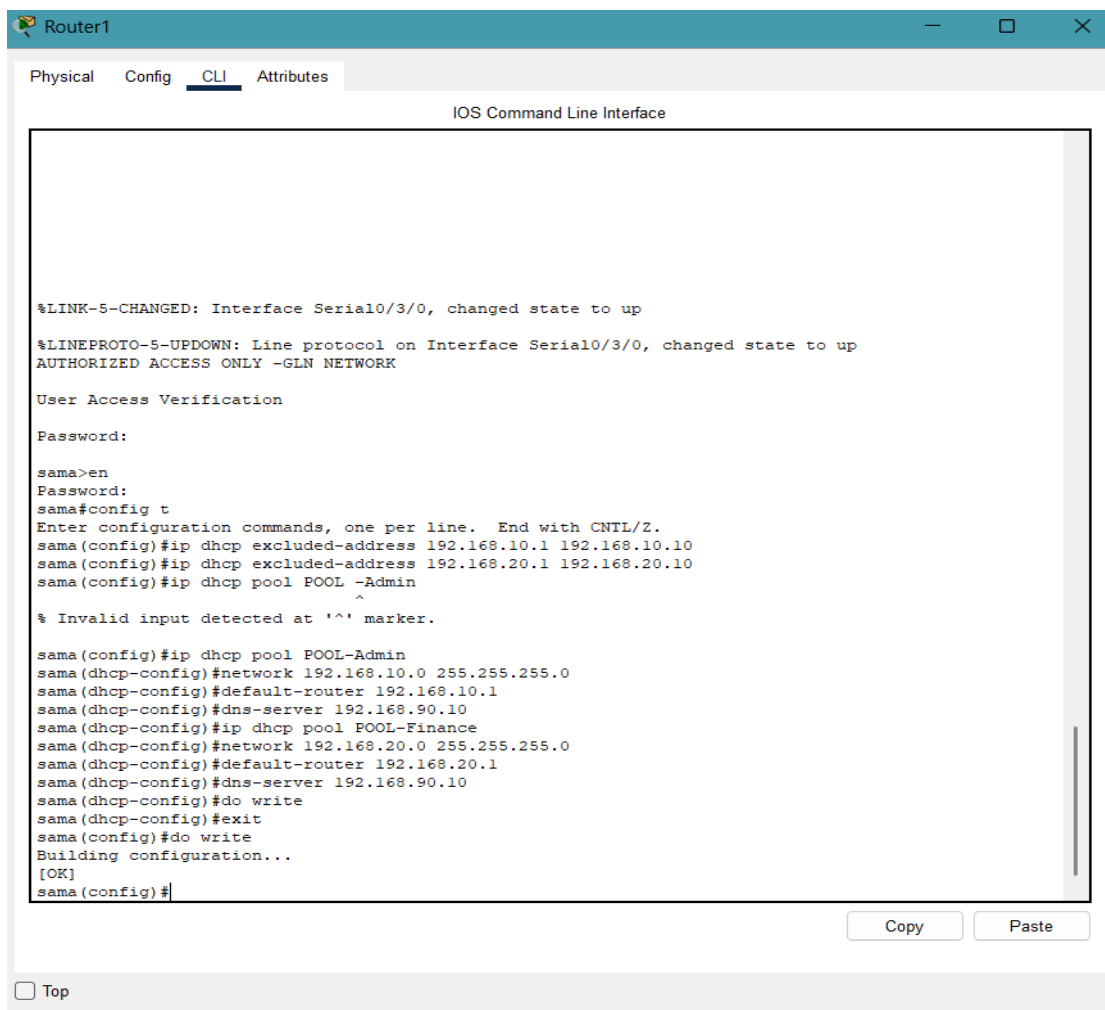
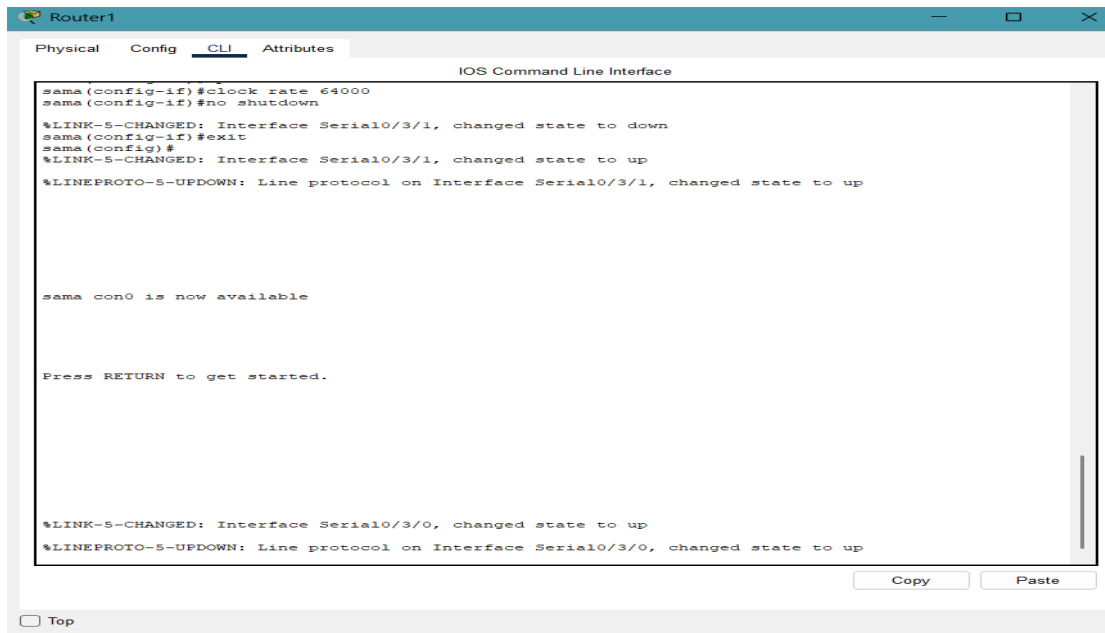
sama(config-if)#interface s0/3/1
      ^
% Invalid input detected at '^' marker.

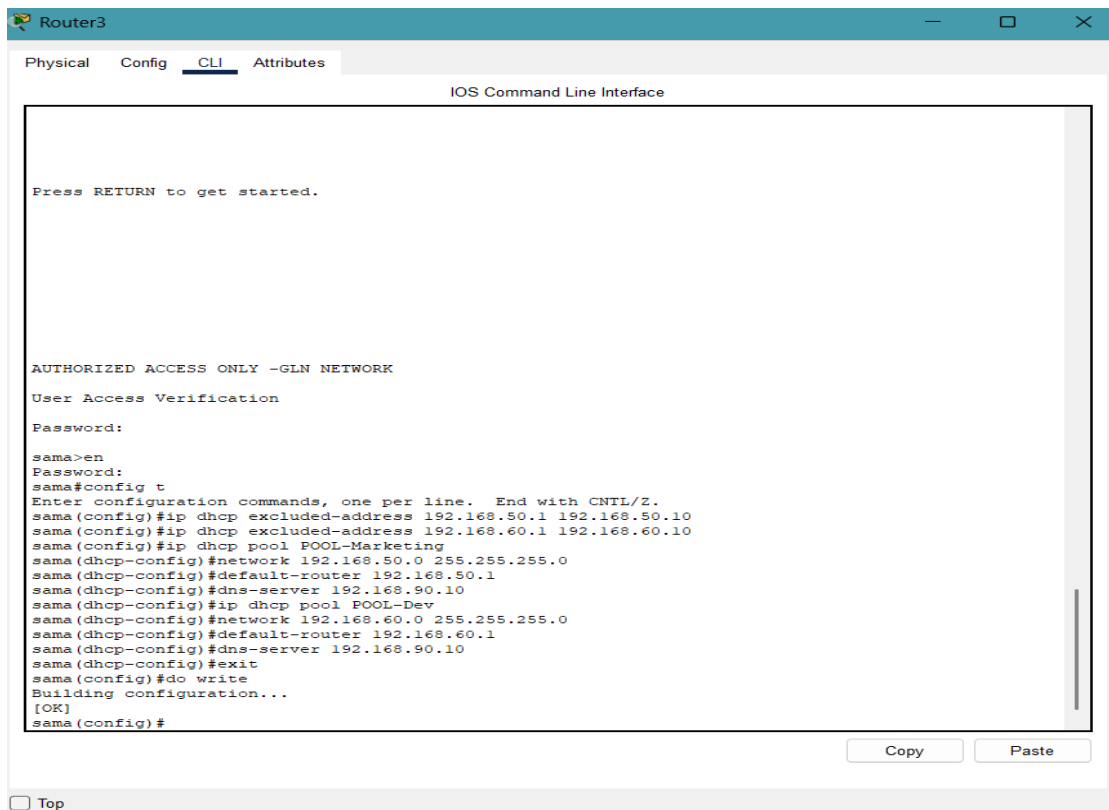
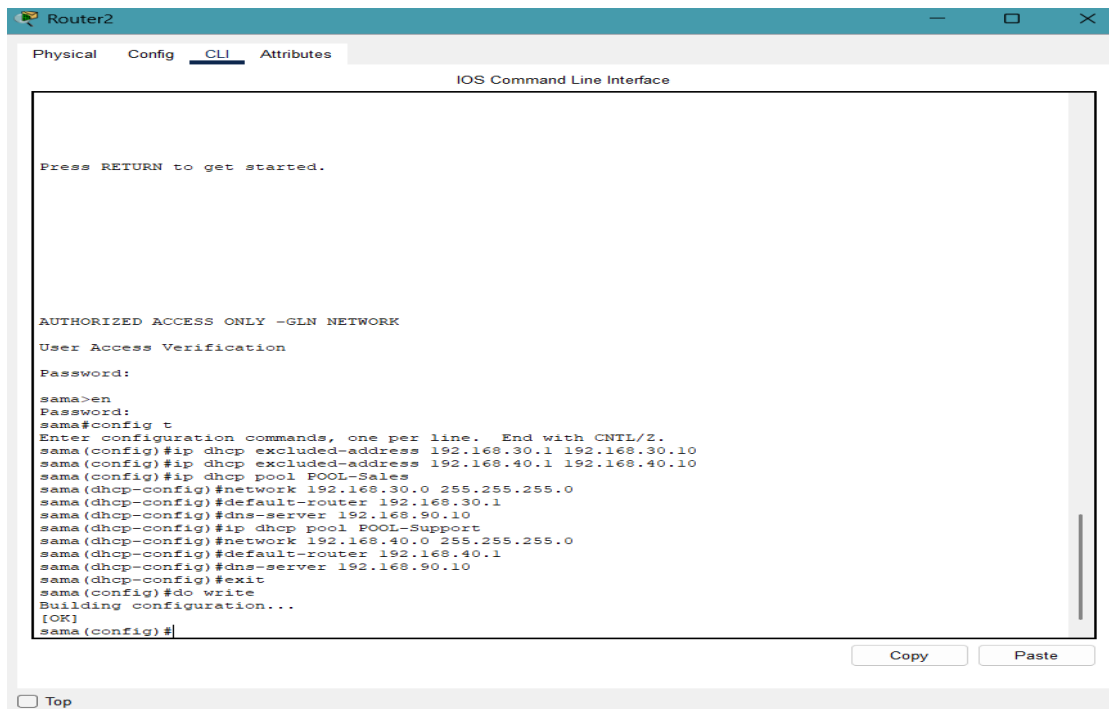
sama(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
interface s0/3/1
sama(config-if)#ip address 10.0.0.17 255.255.255.252
sama(config-if)#clock rate 64000
sama(config-if)#no shutdown

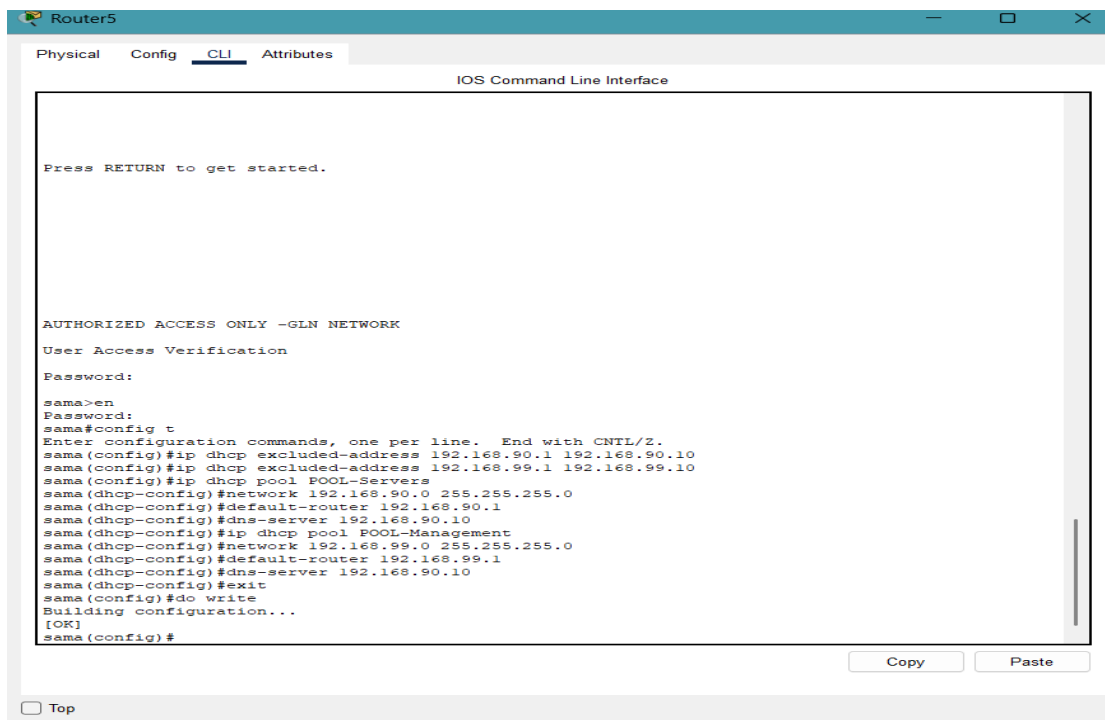
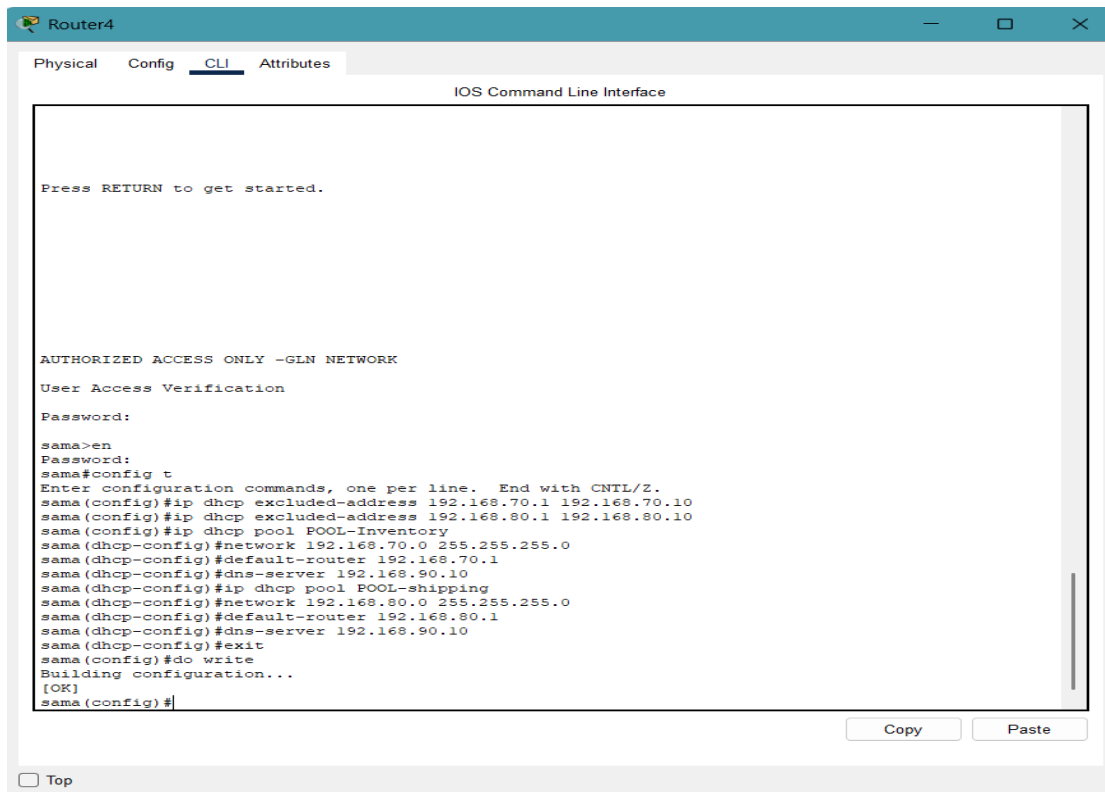
sama(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

sama(config-if)#exit
sama(config)#
```

☐ Top







Router1

PhysicalConfigCLIAttributes

IOS Command Line Interface

Press RETURN to get started.

AUTHORIZED ACCESS ONLY -GLN NETWORK

User Access Verification

Password:

sama>en

Password:

sama#config t

Enter configuration commands, one per line. End with CNTL/Z.

sama(config)#router eigrp 100

sama(config-router)#no auto-summary

sama(config-router)#network 192.168.10.0

sama(config-router)#network 192.168.20.0

sama(config-router)#network 10.0.0.0

sama(config-router)#network 10.0.0.16

sama(config-router)#no network 10.0.0.0

sama(config-router)#no network 10.0.0.16

sama(config-router)#no network 192.168.10.0

sama(config-router)#network 192.168.20.0

sama(config-router)#network 192.168.10.0 0.0.0.255

sama(config-router)#network 192.168.20.0 0.0.0.255

sama(config-router)#network 10.0.0.0 0.0.0.3

sama(config-router)#network 10.0.0.16 0.0.0.3

sama(config-router)#exit

sama(config)#do write

Building configuration...

[OK]

sama(config)#

Copy

Paste

☐ Top

Router2

PhysicalConfigCLIAttributes

IOS Command Line Interface

sama con0 is now available

Press RETURN to get started.

AUTHORIZED ACCESS ONLY -GLN NETWORK

User Access Verification

Password:

sama>en

Password:

sama#config t

Enter configuration commands, one per line. End with CNTL/Z.

sama(config)#router eigrp

% Incomplete command.

sama(config)#router eigrp 100

sama(config-router)#no auto-summary

sama(config-router)#network 192.168.30.0 0.0.0.255

sama(config-router)#network 192.168.40.0 0.0.0.255

sama(config-router)#network 10.0.0.0 0.0.0.3

sama(config-router)#network 10.0.0.4 0.0.0.3

sama(config-router)#exit

sama(config)#do write

Building configuration...

[OK]

sama(config)#

Copy

Paste

☐ Top

Router3

Physical Config CLI Attributes

IOS Command Line Interface

Press RETURN to get started.

AUTHORIZED ACCESS ONLY -GLN NETWORK

User Access Verification

Password:

sama>en
Password:
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#router eigrp 100
sama(config-router)#no auto-summary
^
% Invalid input detected at '^' marker.
sama(config-router)#no auto-summary
sama(config-router)#network 192.168.50.0 0.0.0.255
sama(config-router)#network 192.168.60.0 0.0.0.255
sama(config-router)#network 10.0.0.4 0.0.0.3
sama(config-router)#
%DUAL-S-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.0.5 (Serial0/3/0) is up: new adjacency
sama(config-router)#network 10.0.0.8 0.0.0.3
sama(config-router)#exit
sama(config)#do write
Building configuration...
[OK]
sama(config)#

Copy Paste

☐ Top

Router4

Physical Config CLI Attributes

IOS Command Line Interface

Press RETURN to get started.

AUTHORIZED ACCESS ONLY -GLN NETWORK

User Access Verification

Password:

sama>en
Password:
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#router eigrp 100
sama(config-router)#no auto-summary
sama(config-router)#network 192.168.70.0 0.0.0.255
sama(config-router)#network 192.168.80.0 0.0.0.255
sama(config-router)#network 10.0.0.8 0.0.0.3
sama(config-router)#
%DUAL-S-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.0.9 (Serial0/3/0) is up: new adjacency
sama(config-router)#network 10.0.0.12 0.0.0.3
sama(config-router)#exit
sama(config)#do write
Building configuration...
[OK]
sama(config)#

Copy Paste

☐ Top

The screenshot shows the CLI of Router5. The user has entered the following commands:

```
sama>en
Password:
sama#config t
Enter configuration commands, one per line. End with CNTL/Z.
sama(config)#router eigrp 100
sama(config-router)#no auto-summary
sama(config-router)#network 192.168.90.0 0.0.0.255
sama(config-router)#network 192.168.99.0 0.0.0.255
sama(config-router)#network 10.0.0.12 0.0.0.3
sama(config-router)#
%DUAL-S-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.0.13 (Serial0/3/0) is up: new adjacency
sama(config-router)#network 10.0.0.16 0.0.0.3
sama(config)#do write
Building configuration...
[OK]
sama(config)#
```

Buttons for 'Copy' and 'Paste' are visible at the bottom right of the terminal window.

Finally, I verified the configuration using 'show' commands and saved the running configuration to the startup-config to ensure all changes persist after a device reboot.

The screenshot shows the CLI of Router1. The user has entered the following commands:

```
sama>en
Password:
sama#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

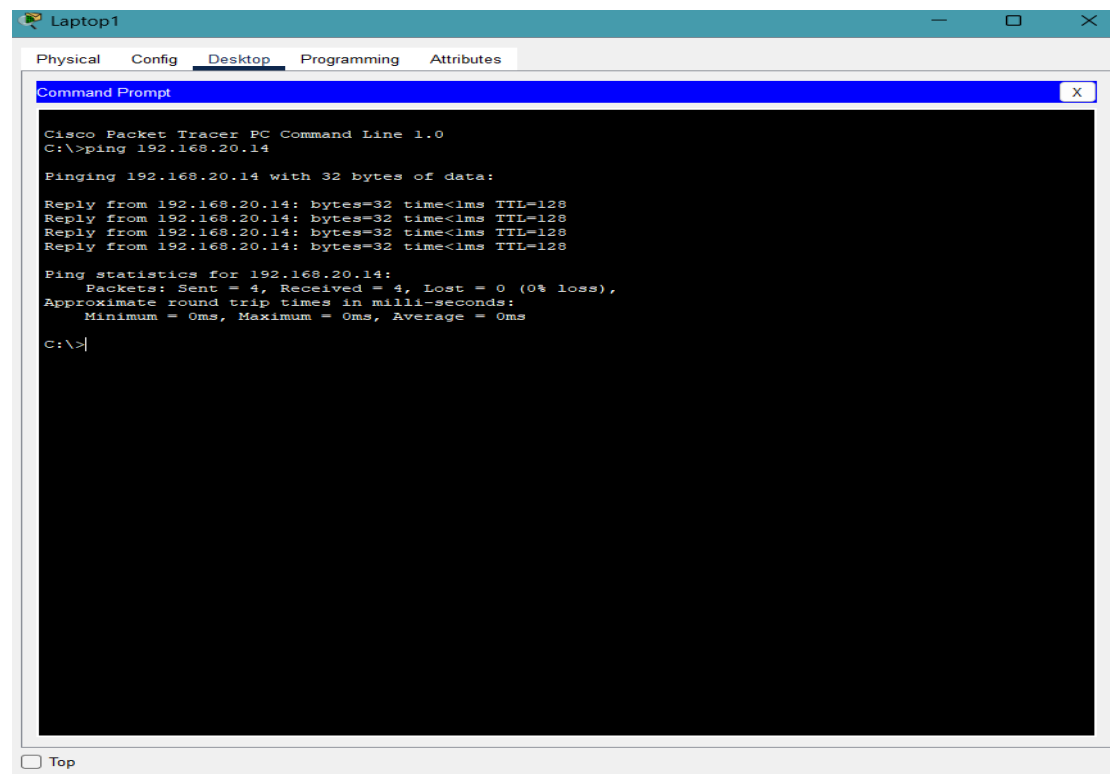
Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C    10.0.0.0/30 is directly connected, Serial0/3/1
L    10.0.0.1/32 is directly connected, Serial0/3/1
D    10.0.0.4/30 [90/2681856] via 10.0.0.2, 01:21:57, Serial0/3/1
D    10.0.0.8/30 [90/3193856] via 10.0.0.17, 01:22:19, Serial0/3/0
     [90/3193856] via 10.0.0.2, 01:21:57, Serial0/3/1
D    10.0.0.12/30 [90/2681856] via 10.0.0.17, 01:22:19, Serial0/3/0
C    10.0.0.16/30 is directly connected, Serial0/3/0
L    10.0.0.18/32 is directly connected, Serial0/3/0
C    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L    192.168.10.1/32 is directly connected, GigabitEthernet0/0.10

sama#show ip interface brief
Interface          IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0 unassigned      YES unset  up              up
GigabitEthernet0/0.10 192.168.10.1    YES manual  up              up
GigabitEthernet0/0.20 192.168.20.1    YES manual  up              up
GigabitEthernet0/1    unassigned      YES unset  administratively down down
GigabitEthernet0/2    unassigned      YES unset  administratively down down
Serial0/3/0          10.0.0.18       YES manual  up              up
Serial0/3/1          10.0.0.1        YES manual  up              up
Vlan1                unassigned      YES unset  administratively down down
sama#
```

Buttons for 'Copy' and 'Paste' are visible at the bottom right of the terminal window.

ping from Labtop1 to Pc1:



The screenshot shows the 'Desktop' tab of a 'Laptop1' device in Cisco Packet Tracer. A 'Command Prompt' window is open, displaying the output of a 'ping 192.168.20.14' command. The output indicates that all four packets were received successfully with 0% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.14

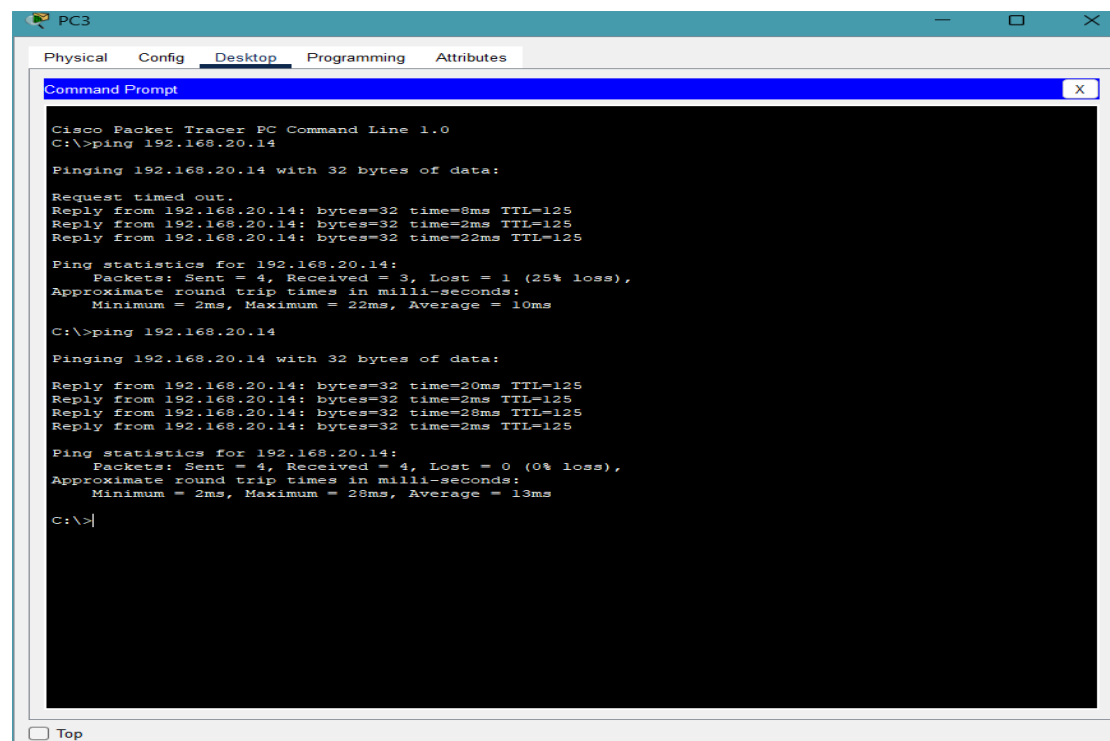
Pinging 192.168.20.14 with 32 bytes of data:

Reply from 192.168.20.14: bytes=32 time<1ms TTL=128
Reply from 192.168.20.14: bytes=32 time<1ms TTL=128
Reply from 192.168.20.14: bytes=32 time<1ms TTL=128
Reply from 192.168.20.14: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

ping from Pc3 to Pc1:



The screenshot shows the 'Desktop' tab of a 'PC3' device in Cisco Packet Tracer. Two 'Command Prompt' windows are shown, both displaying the output of a 'ping 192.168.20.14' command. The first command shows a 'Request timed out' for the first packet and a 25% loss. The second command shows all four packets received successfully with 0% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.14

Pinging 192.168.20.14 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.14: bytes=32 time=8ms TTL=125
Reply from 192.168.20.14: bytes=32 time=2ms TTL=125
Reply from 192.168.20.14: bytes=32 time=22ms TTL=125

Ping statistics for 192.168.20.14:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 22ms, Average = 10ms

C:\>ping 192.168.20.14

Pinging 192.168.20.14 with 32 bytes of data:

Reply from 192.168.20.14: bytes=32 time=20ms TTL=125
Reply from 192.168.20.14: bytes=32 time=2ms TTL=125
Reply from 192.168.20.14: bytes=32 time=28ms TTL=125
Reply from 192.168.20.14: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.20.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 28ms, Average = 13ms

C:\>|
```